

Digital Control

Engineering Technology

May, 2026



Short Title:	Digital Control	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Automotive, Automation and IoT	
Author	CLAIRE.FITZPATRICK	
Credits	5	Level: Advanced

Description of Module / Aims

This course aims to introduce modern techniques used in the analysis and design of digital control systems. These techniques will be demonstrated using examples of practical control systems and will be implemented by the student using MATLAB and SIMULINK. The course should give the student the ability to bring modern control techniques to an industrial setting or perform research in state-of-the-art control methods.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (WD_EAUTO_B)	4 / 8 / M
ENGR-0021	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	4 / 8 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	4 / 8 / M
ENGR-0021	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	4 / 8 / E
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	4 / 8 / M
ENGR-0021	BEng (Hons) (WD_ETRIC_B)	4 / 8 / M
ENGR-0021	BSc (Hons) in Physics for Modern Technology (WD_KPHE_B)	4 / 8 / E

Indicative Content

- Sampling
- Modelling of Sampled Data Systems
- Discrete System Stability Analysis
- Digital Controller Design using Transform techniques
- Direct Digital Controller Design
- Controller Design using State Space Methods
- Three Term Control: analysis and tuning
- Discrete Controller Implementation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Model discrete control systems.
2. Perform stability analysis on discrete systems.
3. Design digital controllers using transform techniques, direct and state space methods.
4. Evaluate controller designs
5. Implement Digital Controller algorithms.
6. Use MATLAB and SIMULINK as part of system analysis and controller design.

Learning and Teaching Methods

- Lectures and Practicals

- - Lectures will involvedelivery of theory and methodology with subsequent problem based learning
- - Practicals will involve use of MATLAB in analysis and design

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	
<i>Practical</i>	20%	4,5,6
<i>Lab Report</i>	10%	4,5,6

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Astrom, K.J. _Computer Controlled Systems; Theory & Design_. 3rd. USA: Dover Publications, 2011.
- Franklin, G.F. _Digital Control of Dynamic Systems_. 3rd ed.. New York: Prentice Hall, 1997.
- Ogata, K. _Modern Control Engineering_. 4th ed.. New York: Prentice Hall, 2001.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics